

EDUGENIE

Solution Architecture

Google Gemini Powered Learning Assistant

Team ID: 6abc95cbdfc814753aa5fe42

Project Name: EduGenie

1. Solution Architecture Diagram

PRESENTATION / CLIENT LAYER
EduGenie Web App — Learning Interface • AI Assistant • Dashboard • Responsive UI
↓ ↑
APPLICATION / CORE LOGIC LAYER
Question Handling • Prompt Processing • Personalized Learning • AI Tutor • Response Generation
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AI / SERVICE LAYER
Google Gemini API • Learning Context • User Preferences • Learning Support
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DATA / DEPLOYMENT LAYER
Session / Progress Data • Web Application Hosting • API Configuration

2. Component Description Table

Component Name	Description / Role in Architecture	Technologies / Services
Presentation Layer	Provides the interface for learner interaction, AI chat, dashboard and personalized learning.	Web UI, Responsive UI
Application / Core Logic	Processes learner questions, prepares prompts and manages the learning interaction.	Application Logic
AI Assistant Layer	Generates answers, explanations and learning assistance.	Google Gemini API
Personalization Layer	Supports personalized learning based on learner needs, preferences and interaction context.	Personalized Learning Logic
Data / Storage Layer	Handles relevant session, preference and progress information where implemented.	Application Storage
Deployment Layer	Hosts and delivers the EduGenie web application.	Web Hosting

3. Architecture Explanation

Client Layer

EduGenie provides a web interface through which learners interact with the learning assistant and receive educational support.

Application Layer

The application receives learner requests, prepares prompts and manages the learning interaction before displaying the generated response.

AI Layer

Google Gemini is the core generative AI service for question answering, explanations and AI-powered learning assistance.

Personalization Layer

EduGenie supports personalized learning using learner needs, preferences and interaction context.

Data Layer

Session, preference and progress information may be maintained according to the implemented application.

Deployment

EduGenie is delivered as a web application accessible through a browser.

4. Architecture Characteristics

Characteristic	EduGenie Implementation
Architecture	AI-powered personalized learning web application
AI Service	Google Gemini
Core Learning Logic	Personalized learning and AI tutor interaction
API	Google Gemini API
Database	Not specified in the supplied project details
Authentication	Not specified in the supplied project details
Hosting	Web application deployment; platform not specified
Project Focus	Google Gemini Powered Learning Assistant

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